⭐⭐⭐⭐⭐ 1️⃣2️⃣ Use a multi-step procedure leading to a mathematical solution

This is the **core competency** assessed.

Students must:

1. Interpret the real-world situation and model it as a geometric figure (a right triangle)
2. Identify the known quantities (opposite side = 150 m, angle = 45°)
3. Identify the unknown quantity (the hypotenuse = rope length)
4. Apply an appropriate mathematical relationship
5. Match the result with the given answer options

➡️ This is a **typical multi-step modelling and problem-solving task**

⭐⭐⭐⭐⭐ 1️⃣3️⃣ Use an understanding of the context to guide or expedite the mathematical solving process

- The word **"approximately"** indicates that:
  - An exact value is not required
  - Estimation or use of special-angle relationships is acceptable
- The angle of **45°** is a highly contextual cue
- Students can reason that the answer:
  - Must be greater than 150 m
  - Should not be excessively large (e.g. close to 300 m)

➡️ Contextual understanding **simplifies and guides** the solution process

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⭐⭐⭐⭐ 1️⃣0️⃣ Make sense of, relate and use a variety of representations

- The problem integrates:
  - Written text
  - Numerical values
  - A geometric diagram
- Students must translate the given height and angle into a geometric relationship

➡️ Successful solution depends on **coordinating textual and visual representations**

⭐⭐⭐⭐ 5️⃣ Understand and utilise constructs based on definitions, rules and formal systems

- Mathematical constructs involved include:
  - Right triangles
  - Properties of a 45°–45°–90° triangle
  - Basic definitions of sine, cosine, or tangent
- These are **standard geometric and trigonometric tools**

➡️ This reflects application of **formal mathematical knowledge**

⭐⭐⭐⭐ 7️⃣ Manipulate numbers, algebraic expressions, and geometric representations

- Students must:
  - Work with the value 150 m
  - Perform multiplication or division (e.g. dividing by sin 45°)
  - Convert geometric relationships into numerical results

➡️ Numerical manipulation occurs alongside geometric reasoning

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⭐⭐⭐ 6️⃣ Develop mathematical diagrams and extract information from them

- The diagram clearly shows:
  - The 45° angle
  - The vertical height
  - The rope as the hypotenuse
- The diagram provides **essential structural information**, not decoration

➡️ Extracting information from the diagram is crucial for modelling

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⭐⭐⭐ 2️⃣ Select an appropriate strategy from a list

Possible strategies include:

- Using trigonometric ratios
- Using the proportional properties of a 45° isosceles right triangle
- Applying the Pythagorean theorem indirectly

Students must select the **most efficient and appropriate approach**

➡️ Strategy selection is present but constrained

⭐⭐ 1️⃣ Perform a simple calculation

- The calculation itself is straightforward (e.g. $150 \div \sin 45^\circ$)
- The challenge lies more in **representation and modelling** than arithmetic

➡️ Calculation is necessary but not the dominant skill

⭐⭐ 8️⃣ Articulate a solution, showing intermediate results

- In a multiple-choice context, full written reasoning is not required
- In instructional settings, intermediate steps may be explained

➡️ This competency is **implicit rather than explicitly assessed**

⭐⭐⭐⭐⭐ 1️⃣2️⃣ Use a multi-step procedure leading to a mathematical solution

（使用多步骤程序得出数学解）

这是本题的**核心能力**：

学生需要：

1. 理解情境并抽象为几何模型（直角三角形）
2. 识别已知量（对边 150 m，角 45°）
3. 确定所求量（斜边 = 绳索长度）
4. 使用合适关系求解
5. 与选项进行近似匹配

➡️ 这是一个**典型的多步骤建模 + 计算问题**

⭐⭐⭐⭐⭐ 1️⃣3️⃣ Use an understanding of the context to guide or expedite the mathematical solving process

（利用情境理解来指导解题，如合适精度）

- “approximately” 表明：
  - 不要求精确值
  - 可接受估算或熟悉角度的简化关系
- 45° 是一个**高度情境化的关键角**
- 学生可判断结果应：
  - 大于 150 m
  - 不会接近 300 m

➡️ 情境显著**降低了计算复杂度**

⭐⭐⭐⭐ 1️⃣0️⃣ Make sense of, relate and use a variety of representations

（理解并使用多种表征）

- 题目结合了：
  - 文字描述
  - 数值信息
  - 几何示意图
- 学生需要将“高度 + 角度”转化为几何关系

➡️ 成功解题依赖于**正确解读图像与文字的结合**

⭐⭐⭐⭐ 5️⃣ Understand and utilise constructs based on definitions, rules and formal systems

（运用定义、规则和熟悉的算法）

- 使用的数学构念包括：
  - 直角三角形
  - 45°–45°–90° 三角形性质
  - 或 sin / cos / tan 的基本定义
- 属于**标准几何 / 三角函数知识的应用**

➡️ 是课程内的**规范数学工具**

⭐⭐⭐⭐ 7️⃣ Manipulate numbers, algebraic expressions and geometric representations

（处理数值和几何表示）

- 学生需要：
  - 操作 150 m
  - 进行乘法或除法（如 ÷ sin45°）
  - 将几何关系转化为数值结果

➡️ 数值处理与几何理解是并行发生的

⭐⭐⭐ 6️⃣ Develop mathematical diagrams and extract information from them

（从图像中提取数学信息）

- 图中明确呈现：
  - 45° 角
  - 垂直高度
  - 绳索作为斜边
- 图像不是装饰，而是**结构性信息来源**

➡️ 图像理解对建模至关重要

⭐⭐⭐ 2️⃣ Select an appropriate strategy from a list

（选择合适的解题策略）

可能的策略包括：

- 使用三角函数
- 利用 45° 等腰直角三角形比例
- 使用勾股定理（通过推理）

学生需选择**最省力、最合理的方法**

➡️ 策略选择存在，但不是开放性的

⭐⭐ 1️⃣ Perform a simple calculation

（进行计算）

- 计算本身并不复杂（如 150 ÷ sin45°）
- 难点不在算，而在**建模与理解**

➡️ 计算是必要但不是主导能力

⭐⭐ 8️⃣ Articulate a solution, showing intermediate results

（表达解题过程）

- 在选择题中，通常不要求书写完整推导
- 但若在教学或解释中，需要展示中间步骤

➡️ 属于**潜在能力，而非显性考查**